



Research on soil structure analysis and VR game experience of Ocean city based on image resolution

Zijian Li¹ · Xiao Yu²

Received: 31 May 2021 / Accepted: 30 June 2021 / Published online: 3 August 2021
© Saudi Society for Geosciences 2021

Abstract

After 1950, *Casuarina equisetifolia* was introduced into coastal areas as a windbreak forest and sand-fixing forest, adjusted the balance of the ecosystem, and played an irreplaceable role in resisting disasters and protecting coastal land. *Casuarina equisetifolia* is the main tree species in coastal protected forests in Fujian Province. However, due to long-term pure forest management and human intervention, the acidification of forest land is serious, soil nutrients are unbalanced, soil fertility is reduced, reforestation becomes difficult, and the stability and sustainable management of coastal windbreak forests are insufficient. Therefore, forestry in a certain region chose the contrast of bare sand and used it as the research object to study the physical and chemical properties of soil, and to study soil enzyme activity in pure casuarina forest, mixed forest of *Casuarina equisetifolia* and Ginger ginger, and natural secondary the vegetation restoration model in the three cases of forest. Investigate the soil microbial characteristics under different restoration modes, and the degraded sandy soil properties under different vegetation restoration modes. This research is based on the theoretical basis of restoring vegetation on the deteriorating sandy land along the southeast coast, and provides a theoretical basis for the construction of coastal protection belts for natural forests. In recent years, virtual reality (VR) technology has been applied in various fields, achieving large-scale commercialization. So far, the research of VR game user experience has mainly focused on usability, VR technology research, effect analysis, immersion and so on. Maintaining user mobility will help optimize user experience and improve VR game design.

Keywords Image resolution · Ocean city · Soil structure · VR game experience

Introduction

After 1950, *Casuarina equisetifolia* was introduced into coastal areas as a windbreak and sand-fixing forest to adjust the ecological system balance in the area, improve environmental quality, resist tsunamis and sandstorms, and protect the safety of coastal land and ecosystems (Ali 2016). It plays an important role in the whole process: the role of substitution.

However, in the early stage of afforestation, *Casuarina equisetifolia* forests are basically pure forests because of the single composition of seeds, simple structure of windbreak forests, low stability, and frequent occurrence of diseases and pests (Bai et al. 2020). Due to forest soil acidification, malnutrition, reduced soil fertility, difficulties in replanting afforestation, reduced wind-proof effect, and insufficient stability and sustainable management of coastal wind-break forests (Cuderman 1986). In recent years, in a certain region, especially during the “Twelfth Five-Year Plan” period, the construction of coastal windbreak forests has been increased. A lot of human resources and funds have been invested in the construction of coastal windbreak forest systems (Feng et al. 2018). However, soil problems still exist such as acidification, nutrient imbalance, and fertile soil degree drops. In the sandy soil along the southeast coast, the research on coastal trench forests mainly focuses on the vegetation characteristics, physiological ecology, remote sensing monitoring, upgrading technology, and soil physical and chemical properties of coastal trench forests, accumulating rich practical experience, and the

This article is part of the Topical Collection on *Environment and Low Carbon Transportation*

Responsible Editor: Sheldon Williamson

✉ Zijian Li
jiuwoshujiu81788@163.com

¹ School of Fine Arts, Central China Normal University, Wuhan 430079, Hubei, China

² School of Arts, Hubei University of Education, Wuhan 430205, Hubei, China

new theory of soil restoration, but there is still a lack of research on soil microorganisms in windbreaks and sand-fixing forests (Gao et al. 2017). Therefore, it is very important to study the soil characteristics and microorganisms of vegetation restoration in barren sandy land along the coast (Han 1994).

Virtual reality games are an active entertainment industry. Games are also one of the favorite entertainment items of young people (Hossain et al. 2000). From the point of view of the game experience, the user experience represents a complex process that uses a variety of components and measurement methods to be evaluated in many ways by many researchers (Huang et al. 2014). During the game, people may experience an all-round deep experience called flow. Flow resources are an important factor in virtual reality (VR) user resources (Huang et al. 2018). If the player has a mental state when playing the game, he can not only devote himself to the game, but also enjoy the fun of the game, thus ignoring the changes of the outside time. Flow is a state of mind for people to enjoy activities (Jaimes et al. 2012). This state is seen by players, is an ideal experience for players, and is a research direction that cannot be ignored for game designers. With the rise of VR games, the study of flow experience and related elements is an important research field (Khabirov 2016).

Materials and methods

Plot setting and survey methods

Typical *Casuarina equisetifolia* pure forests, *Casuarina equisetifolia* + ginger ginger mixed forests, and natural secondary forests are selected locally. Three 20 m × 20 m maps were set up for each treatment, and the sampling area was 1200 m². Soil evaluation was carried out in the district in December 2019, and a community survey was carried out in May 2020. Table 1 shows the basic information of the two sample regions.

Soil collection method

As shown in Fig. 1, soil samples were collected according to the “5-point” mixed sampling method. For each sample map, we dug 5 soil pits with a depth of 60 cm, divided into three

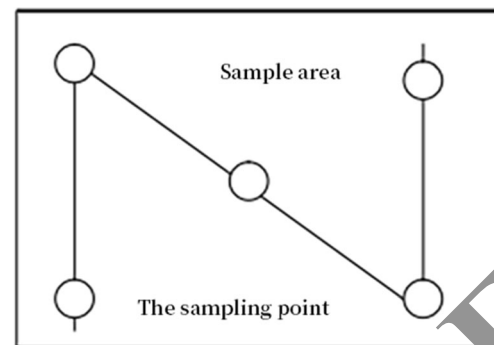


Fig. 1 Schematic diagram of mixed sampling

types: 0–20 cm, 20–40 cm, and 40–60 cm sampling. Samples of physical properties are collected from top to bottom, and samples of chemical properties and soil microorganisms are collected from bottom to top. Remove plant roots and sand and gravel in the soil and mix the soil of the soil layer at an equal ratio to form a mixed sample. Divide into 2 parts using the method of 4 points, put one of them into the sterilization bag immediately, then put it into the sampling box with ice, filter through 2-mm sieve, and store it in the refrigerator at 4 °C (below), but time must be less than three days, and this part of the sample is used for soil microbial analysis. Another part of the soil samples were naturally dried and crushed indoors in order to determine the activity and physical and chemical properties of soil enzymes, and screened with 1-mm and 0.15-mm sieves.

Soil structure analysis method

Determination methods of soil physical and chemical properties

The physical and chemical properties of the soil are determined using the methods in the “People’s Republic of China Forestry Industry Standard.”

- ① Soil bulk density: ring knife method;
- ② Soil water content: drying method;
- ③ Soil pH: electric potential method;
- ④ Total soil nitrogen: semi-micro Kjeldahl method;
- ⑤ Soil hydrolyzable nitrogen: alkaline diffusion method;
- ⑥ Total phosphorus in soil: alkali fusion-molybdenum antimony colorimetric method;

Table 1 Basic situation of survey samples

Sample number	Slope/°	Aspect	Altitude /m	Canopy closure /%	Dominant species	Soil	Tree height/m
APF	10–15	North	40	90	Casuarina	Types of	18
CK	10–12	North	20	0	No	Aeolian	0

- ⑦ Total potassium in soil: alkali fusion-flame photometry;

Determination method of soil enzyme activity

The determination method of this study adopts the method in “Soil Enzymes and Research Methods.”

- ① Urease: using phenol-sodium hypochlorite colorimetric method;
- ② Catalase: KMnO₄ titration method;
- ③ Polyphenol oxidase: iodometric titration method.

Methods for determining the quantity of soil microorganisms

Number of bacteria per gram of dry soil

$$= (\text{average number of colonies} \times \text{dilution factor}) / (\text{percentage of dry soil}) \quad (1)$$

It is usually expressed as cfu (colonyformingunit)/g dry soil.

Number of bacteria per gram of dry soil

$$= (\text{average number of colonies} \times \text{dilution factor} \times 100) / (\text{percentage of dry soil}) \quad (2)$$

It is usually expressed as cfu (colonyformingunit)/g dry soil.

Methods for determination of soil microbial biomass

$$MB = F/K \quad (3)$$

F is the difference between the fumigated sample and the unfumigated sample, and K is the conversion factor.

Design of super-resolution reconstruction algorithm for remote sensing image

Sparse representation models in compressed sensing usually work in a block-based framework, where images are redundantly represented by the sparse coefficients of their overlapping blocks. Then, the restoration of the image will average the sparse representation of all overlapping blocks. Considering the original image x in the noise observation model y , the mean value of w is 0, which can be expressed as formula (4):

$$y = x + w \quad (4)$$

A commonly used reconstruction method is to obtain the sparse decomposition of the image block by maximizing and minimizing the problem, as shown in formula (5):

$$\hat{\alpha}_i = \arg\min_{\alpha} \frac{1}{2} \|y_i - D\alpha\|_2^2 + \mu \|\alpha\|_p \quad (5)$$

Given a set of training image blocks $\{x_i\}_{i=1}^N$, the learned sparse dictionary is usually expressed as shown in formula (6):

$$\min_{D \in D, \{\alpha_i\}_{i=1}^N} \sum_{i=1}^N \left(\frac{1}{2} \|x_i - D\alpha_i\|_2^2 + \mu \|\alpha_i\|_p \right) \quad (6)$$

This paper uses the representation of non-locally similar blocks to generate a better estimated image of the original block, where the local sparse regularization term is replaced by a non-local regularization term as shown in formula (7):

$$\hat{\alpha}_i = \arg\min_{\alpha} \frac{1}{2} \|y_i - D\alpha\|_2^2 + \mu \|\alpha - k_i\|_1 \quad (7)$$

The i -th block of k_i can be defined as shown in (8):

$$k_i = \sum_{j \in \Omega_i} w_{ij} \alpha_j^{(0)} \quad (8)$$

The weight w_{ij} depends on the similarity between the i -th block and other blocks as shown in formula (9):

$$W_{i,j} = \frac{1}{A_i} \exp\left(-\frac{\|y_i - y_j\|_2^2}{h}\right), j \in \Omega_i \quad (9)$$

Statistics and processing of data

Data statistics and processing are completed in Excel 2003 and SPSS 17.0.

Results

Soil physical and chemical properties of different marine urban sand restoration models

Bulk density is an important index to characterize soil structure. If the soil is similar, the bulk density reflects soil particles, soil permeability, aeration, and root resistance. The bulk density of soil mainly depends on the soil structure, which has a great influence on the activities of tree roots, soil fauna, and microorganisms. It can be seen from Fig. 2 that natural secondary forests, mixed forests, pure forests, and bare sandy land gradually increase bulk density from top to bottom, but

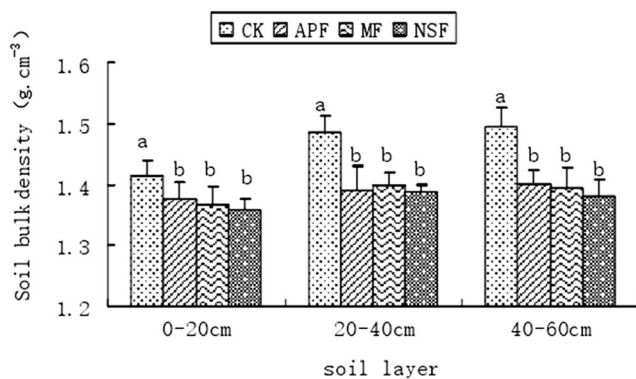


Fig. 2 Soil bulk density of different restoration modes

the increasing trends are different, and changes in soil density make various vegetation restoration models different (Kutter and Fairhurst 1971). The change trend of soil roughness density after vegetation restoration is as follows: natural secondary forest < Mixed ginger + *Casuarina* forest < Pure *Casuarina* forest, especially the soil layer with a bulk density of 0–20 cm in the soil layer of 0–20 cm. Compared with the corresponding bare sand land, it is reduced by 9.03%, 1.23%, and 0.51% respectively (Lanari and Fakhimi 2015). The bulk density of the pure *Casuarina equisetifolia* forest is the largest, but the bulk density of the mixed plantation of ginger and *casuarina equisetifolia* is reduced compared with that of bare sand. Compared with the single species of *Casuarina*, it has a greater impact on rainfall closure, so the surface hardness is less and the density is also low (Li et al. 1995). The three cultivation and vegetation restoration modes all show that the umbrella density of the soil is reduced, and vegetation restoration can improve the soil structure.

Soil moisture is an important factor for plant growth and development. The dynamic balance is produced by the combined effect of the water content, water cycle, and water balance in its spatial distribution area. Figure 3 shows the order of soil moisture content as follows: natural secondary forest > mixed forest > pure forest > bare sandy land. After the restoration of vegetation, the water quality required for vegetation growth can be improved, thereby improving the ecological environment of the sandy land. Changes in soil moisture content fully reflect the surface vegetation.

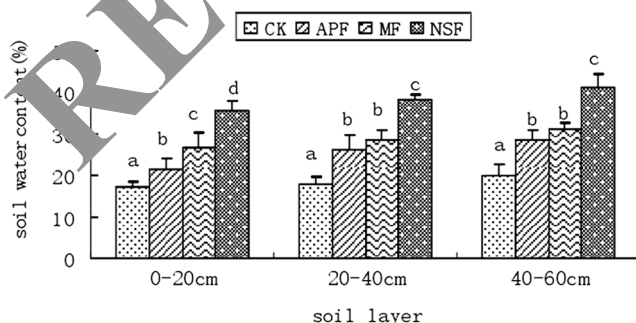


Fig. 3 Soil moisture content of different restoration models

Restoring vegetation can increase soil moisture, and adding surface coating can prevent the evaporation of soil surface moisture. This reflects the positive effect of vegetation restoration on soil moisture. The water content of natural secondary forests is significantly higher than that of pure forests and mixed forests. Natural secondary forests can effectively maintain soil moisture because of their complete cluster structure and a rich recirculation layer on the surface. However, the artificially restored mixed forest is higher than the pure forest, and the difference in the surface layer is very large, and the difference will be small as the depth increases. The bulk flow of the surface layer has a great impact on the surface soil moisture.

From the point of view of the soil's ability to retain water, the water in the capillary pores can be retained in the soil for a long time, mainly for the absorption of plant roots and soil evaporation. It can be seen from Table 2 that the soil water retention capacity under different restoration modes is higher than that of the corresponding bare sand land (Liu and Zhao 2008). The minimum capacity is 7.08, 5.29, and 4.90 percentage points higher than the comparison group, and the increase rates are 60.55%, 41.20, and 41.89% respectively (Liu et al. 2018). From the perspective of the vertical soil structure, the natural secondary forest after natural restoration has the largest water retention capacity (Ma and An 2008). Compared with other restoration methods, it has stronger water protection and water saving capabilities, and can reduce the outflow of surface water caused by rainfall intensity (Majid et al. 2015).

Among the various restoration models, only the total void ratio of the natural secondary forest exceeds 40%. The other two models are not within a good range, indicating that the number of non-capillary pores in the pure forest of *Casuarina* is very small (Passamaneck 1993). The permeability and water retention are very weak. Therefore, improving the non-capillary pores can more effectively improve the pore structure. Regarding the improvement value and improvement rate of soil porosity under different restoration modes, the natural secondary forest restored by nature has the strongest improvement effect on the soil porous structure. Please refer to Table 3.

The change of soil pH directly affects the state, exchange, movement, and transformation of ions in the soil, and also affects the conversion of soil materials, microbial activities, the possibility of nutrient utilization, and soil characteristics, and plays an important role, especially in soil the formation of fertility and the evolution of soil quality. After vegetation restoration on bare sandy land, the pH value showed a downward trend (Rao et al. 2014). Among them, the pH value of mixed forest and pure forest decreased by 29.87% and 44.72% compared with bare sand land, and the acidification of planting soil was serious (Rossmanith et al. 1996). The pH value of the natural secondary forest is higher than the pH value of the artificially restored mixed forest and pure forest. The pH value

Table 2 Soil water holding capacity of different restoration models

Sample number	Soil layer	Maximum water holding capacity	Capillary water holding capacity	Minimum water holding capacity
NSF	0–20	26.741	23.714	19.838
	20–40	22.733	20.537	19.050
	40–60	23.820	21.305	17.455
MF	0–20	21.862	20.659	18.943
	20–40	20.637	18.234	16.383
	40–60	19.255	15.875	15.630
APF	0–20	21.785	18.292	17.838
	20–40	20.633	16.899	16.615
	40–60	19.241	16.125	15.233
CK	0–20	17.212	14.382	12.153
	20–40	16.968	13.628	11.908
	40–60	16.348	12.317	11.036

of the natural secondary forest was 9.68% higher than that of the mixed forest, 22.2% higher than that of the pure forest, and 18.41% lower than that of the bare sandy land. Therefore, it can be known that the bare sand land after vegetation restoration, the vegetation restoration is good, and the decrease in soil pH is small. This is due to the impact of the accumulation of organic acids in the soil on the vegetation and soil pH, and the impact on the vegetation is the absorption and utilization of nitrogen and phosphorus in the soil, and the composition of roots and root exudates (Sahil et al. 2017). Soil pH increases with the increase of soil depth under various restoration modes (as shown in Fig. 4).

It can be seen from Table 4 that the soil organic matter content in the 3 planting restoration mode has increased significantly. The content of natural secondary forest is the highest, which is 3.3355 g Kg⁻¹, while bare sand land is only 1.013 g Kg⁻¹. In vertical soil distribution, organic matter decreases as the soil depth increases. In the 0–20 cm soil layer,

the soil organic matter content of bare sand land is 1.289 g Kg⁻¹, and the reduction rates are 12.82%, 7.50%, 42.13%, and 69.65%, respectively. In contrast, the increase in organic matter is mainly in the 0–20 cm soil layer, and gradually decreases in the 40–60 cm soil layer. The main reason is the natural restoration of plant diversity in the natural secondary forest, which increases the complexity of the tree structure, provides a rich carbon source for soil microorganisms, and can be quickly decomposed, thereby speeding up the circulation of organic matter and promoting the growth and development of plants.

Analysis of soil microbes and soil quality in different marine urban sandy land restoration models

Soil microorganisms are an important element of the forest ecosystem, involving the material and energy cycle of the system. The quantity, distribution, and activities of

Table 3 Soil porosity of different restoration models

Sample number	Soil layer	Non-capillary porosity	Capillary porosity	Non-capillary porosity/capillary porosity	Total porosity
NSF	0–20	13.795	31.372	1:2.25	45.164
	20–40	12.482	29.574	1:2.33	42.055
	40–60	10.256	29.097	1:3.00	39.354
MF	0–20	8.288	21.796	1:2.67	30.085
	20–40	7.624	22.353	1:3.00	29.976
	40–60	6.276	21.898	1:3.50	28.175
APF	0–20	5.174	26.945	1:5.00	32.118
	20–40	4.827	24.570	1:5.00	29.398
	40–60	3.919	27.528	1:7.00	31.446
CK	0–20	2.988	19.311	1:6.00	22.297
	20–40	2.371	18.288	1:8.00	20.661
	40–60	1.965	18.733	1:9.00	20.295

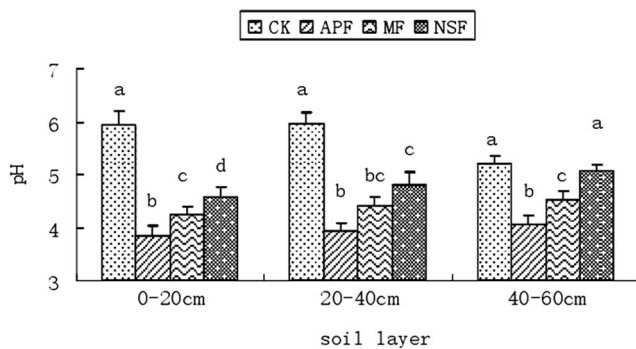


Fig. 4 Soil pH value of different recovery modes

microorganisms in the soil are affected by plant roots, soil fertility, soil moisture, temperature, and other factors, which play an important role in the growth and development of plants. It can be seen from Fig. 5 that no matter how the number changes, the main trends are bacteria > actinomycetes > fungi > aerobic self-nitrogen-fixed bacteria. Under various recovery modes, the order of the total number of bacteria, fungi, actinomycetes, and microorganisms is as follows: natural secondary forest > mixed forest > pure forest > bare sand. The aerobic self-generating nitrogen-fixing bacteria are natural secondary forest > pure forest > mixed forest > bare sandy land.

It can be seen from Table 5 that the carbon, nitrogen, and phosphorus content of microbial biomass in bare sand is the lowest, which are 15.9 mg kg^{-1} , 3.118 mg kg^{-1} , and 0.943 mg kg^{-1} , respectively. Compared with bare sandy land, the biomass of soil microbes under different recovery modes increased significantly ($P < 0.01$), and the increase rate of carbon, nitrogen, and phosphorus of soil microbial biomass was 103.6966–491.166%, 89.352–335.087%, and 15.536–88.896%. The phosphorus content of microbial biomass under different recovery modes is as follows: NSF > APF > MF, which is 1.859 times, 1.852 times, and 1.555 times that of bare sand. In the sandy soil along the coast of D County, the proportion of minerals released in the sands is very limited, and

microbial biomass has a great influence on the conversion and circulation of C, N, and P in the soil. After vegetation restoration in bare sand soil, the increase of soil microbial biomass will increase soil nutrients on the one hand, and improve the utilization of soil nutrients on the other hand, making soil conditions more conducive to plant growth.

A line chart was made according to the IFI value, and the relationship between the quality of soil fertility and vegetation restoration under various restoration modes was further analyzed. As shown in Fig. 6, the soil fertility under different restoration modes is 0–20 cm layer > 20–40 cm layer > 40–60 cm layer, indicating that soil fertility decreases as the soil level deepens. The quality of different fertility of soil fertility is as follows: natural secondary forest > hybrid forest > pure forest > bare sandy land. In bare sand, there are no trees and shrubs, few herbs, no bioaccumulation in the soil, and poor soil quality. After the pure forest was restored, the vegetation improved the state and quality of the soil (Sun et al. 2010). Due to the increase of broad-leaved trees, the number of shrubs and herbs in the area increased, enzyme activity increased, soil nutrient circulation accelerated, and soil fertility increased significantly. Due to changes in tree structure, natural secondary forests have increased the number and types of broad-leaved tree species, and the number and types of shrubs and herbs have also increased sharply, resulting in richer biodiversity and stronger soil fertility quality capabilities (Sun et al. 2017). More soil animal microbes and types have improved the activity of soil enzymes and the quality of soil has been further improved.

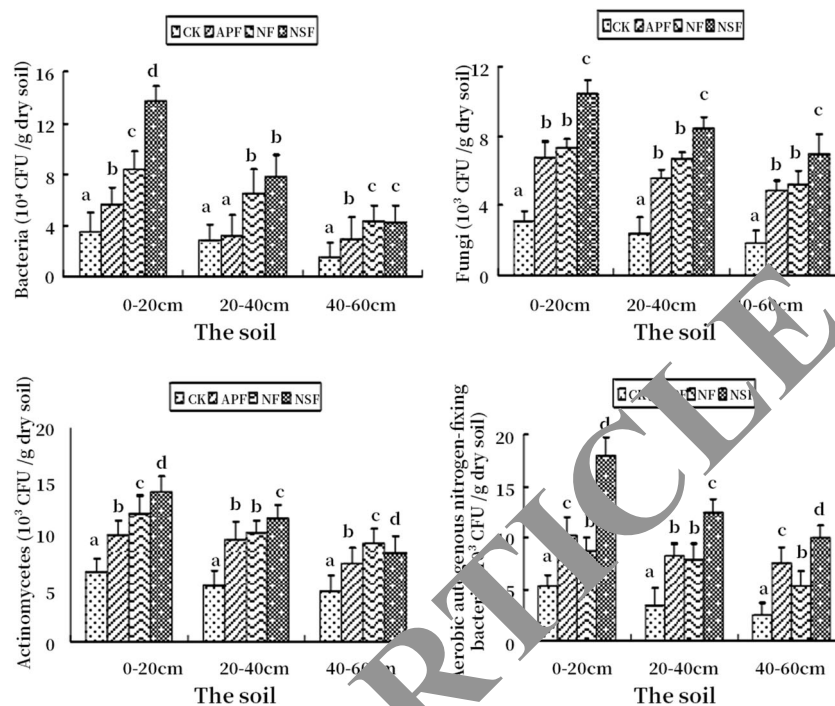
Analysis of soil microbial community structure in marine cities

Figure 7 shows the colony morphology of some representative bacterial strains.

Table 4 Soil nutrients of different restoration models

Sample number	Soil layer	SOM	TN	HN	TP	AP	TK	AK
NSF	0–20	5.652	0.727	34.518	0.163	0.595	16.278	43.791
	20–40	3.386	0.355	16.683	0.138	0.435	15.383	32.324
	40–60	1.027	0.296	12.390	0.118	0.215	13.581	24.683
MF	0–20	3.518	0.468	28.835	0.144	0.355	12.371	32.427
	20–40	1.793	0.325	17.275	0.127	0.245	11.425	28.611
	40–60	1.038	0.214	15.760	0.104	0.256	10.895	21.273
APF	0–20	2.938	0.396	12.368	0.135	0.267	11.306	17.894
	20–40	1.348	0.162	9.493	0.120	0.136	10.744	15.583
	40–60	0.979	0.144	5.370	0.100	0.245	10.186	12.647
CK	0–20	1.287	0.193	7.524	0.104	0.157	8.537	11.723
	20–40	0.965	0.168	5.377	0.094	0.145	7.822	10.026
	40–60	0.845	0.178	4.216	0.094	0.125	7.058	9.218

Fig. 5 The number of microorganisms in different recovery modes



In order to improve efficiency, in the array determination process, multiple samples are usually determined at the same time. In other words, multiple samples are mixed to determine the arrangement. In order to distinguish between different samples, a barcode sequence label indicating the source of the sample must be added before each sample sequence. Then, the data statistics of the effective sequence is to delete the barcode and the forward main sequence, and perform the data and length distribution statistics of the effective sequence after adoption, classification, and processing. Please refer to Figs. 8 and 9 for details.

Dilution curve refers to randomly selecting a specific number of individuals from a sample, counting the number of species represented by these individuals, and creating a curve

based on the number of individuals and species. Using this option, you can compare the richness of multiple sample types with different sequence sizes and show whether the sample size is appropriate.

Numsampled: The number of random samples (decided in advance to start from 1, this sample will be recalculated every time it increases by 100), as shown in Fig. 10.

It can be seen from the experimental results that when the similarity of OTU is 90% (0.10) (blue line in the figure) and 95% (0.05) (black line in the figure), the number of OTUs will gradually increase. In order to obtain a more accurate cluster analysis, a similar level of OTU of 97% (0.03) (the red line in the figure) was selected in the drawing. If the sequence

Table 5 Microbial biomass in different recovery modes

Sample number	Soil layer	MB _C	MB _N	MB _P	MB _C /MB _N	MB _C /SOM	MB _N /TN	MB _P /TP
NSF	0–20	151.844	20.280	2.318	6.005	2.688	3.474	1.432
	20–40	90.246	8.435	1.675	10.702	2.666	2.383	1.221
	40–60	40.978	6.983	1.268	5.868	3.987	2.350	1.065
MF	0–20	79.918	14.257	1.764	5.606	2.270	3.041	1.234
	20–40	45.647	8.868	1.614	5.146	2.546	2.713	1.261
	40–60	35.198	6.143	1.307	5.732	3.395	2.856	1.272
APF	0–20	52.279	10.884	2.300	4.805	1.782	2.740	1.693
	20–40	26.604	3.633	1.574	7.326	1.971	2.255	1.301
	40–60	18.783	3.199	1.363	5.876	1.922	2.235	1.351
CK	0–20	20.426	3.419	1.042	5.975	1.584	1.781	0.994
	20–40	13.934	3.129	0.937	4.455	1.440	1.851	0.986
	40–60	13.524	2.812	0.849	4.813	1.605	1.571	0.913

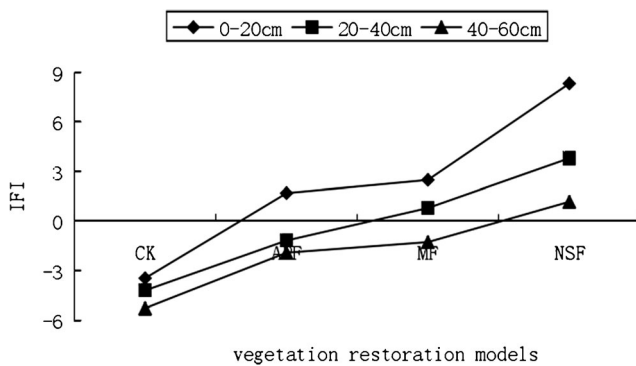


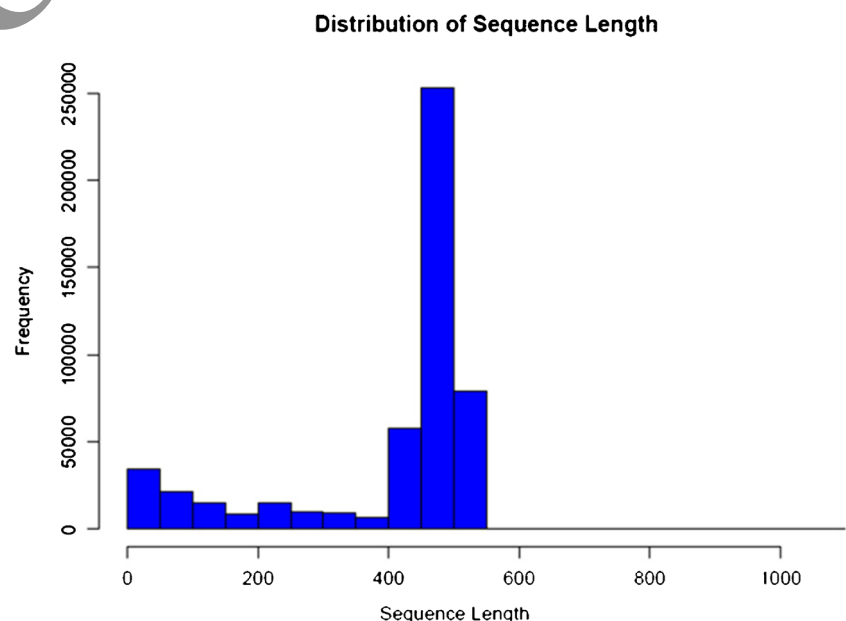
Fig. 6 Comprehensive evaluation of soil indicators for different restoration models



Fig. 7 Colony morphology of some representative bacterial strains

exceeds 35,000, the OTU reaches 7000, and there is no obvious upward trend. This indicates that the sample library has a high coverage rate, and the sequence results can indicate the actual situation of the sample.

Fig. 8 The length distribution of each sample of the effective sequence



In this analysis, OTU splits the level with a similarity of 97% to create a sparse curve for each sample, as shown in Fig. 11.

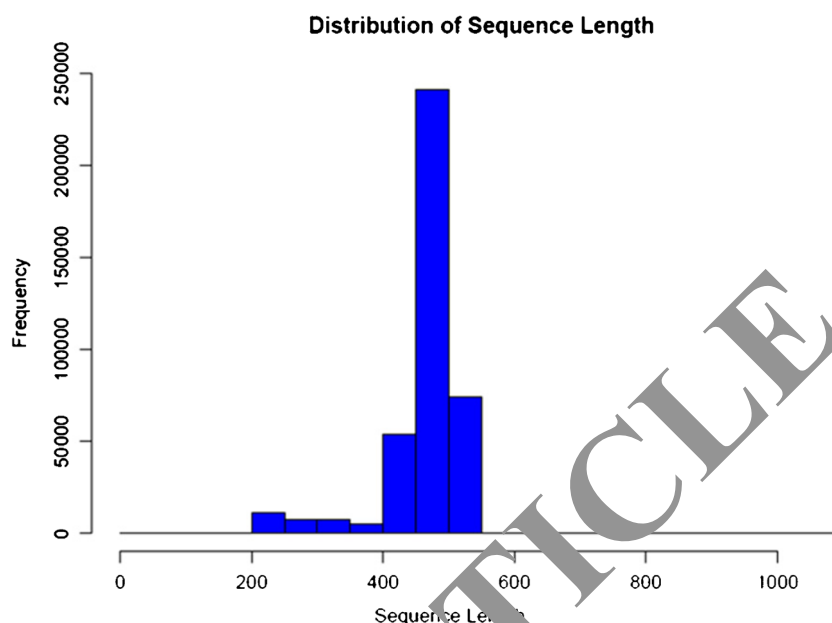
At the taxonomic level, a bar graph was made based on the microbial composition of each sample, and the differences in group composition at the taxonomic level of each sample were compared, as shown in Fig. 12.

The number of common OTUs of multiple samples may reflect the similarity and repeatability of environmental samples. The statistical results are obtained in the form of a curved graph. The similarity level of OTU used in this analysis is 97% (0.03). Group 1 is located in Arctic Salix soil (site 1, 2, 3), Group 2 is located in Arctic rhizosphere soil (site 4, 5, 6), Group 3 is located in the rhizosphere soil of Polygonaceae plants (sites 7, 8, 9), and Group 4 is located at the background level (sites 10, 11, 12). The analysis results are shown in Fig. 13.

It can be seen from this figure that in the 40861 sequence, only the 788 sequence is the common sequence of the four regional soils, which shows that different regions have different characteristics and the number of sequences between each station and the background is similar. Group 1 and Group 2 have obvious similarities. Then, the data is often grouped according to needs. The similarities and differences of the clustered data are shown in Fig. 14, which is reflected in the degree of color gradation and similarity shown in the heatmap.

The results of PCA analysis are usually represented by two-dimensional or three-dimensional dot plots. Generally speaking, a two-dimensional bitmap is convenient for observation and analysis, but if the two-dimensional results cannot distinguish the samples, a three-dimensional bitmap as shown in Fig. 15 can also be used.

Fig. 9 Optimized sequence data length distribution



Combining the physical and chemical properties of the soil with the characteristics of different stations for principal component analysis is shown in Fig. 16.

Discussion

Soil properties of coastal shelter forests

Many studies have been conducted on the physical and chemical properties of the soil in coastal windbreak forests at home and abroad. MailyD studied the *Casuarina* forest on the coast

of Senegal and pointed out that after setting up windbreaks, the soil pH dropped by 12.3%, the thickness of deciduous branches increased by 90.48%, and the net biomass of Ca, Al, Mg, Fe, K, and P in the vegetation increased significantly. As the forest age increases, nitrogen and organic matter also increase year by year.

In recent years, many domestic scholars have begun to study the soil enzyme activity and rhizosphere soil of windbreak forests. Through research on the relationship between soil enzyme activity and physical and chemical properties of the mud-filled coastal trenches, Hu Haibo said that in the

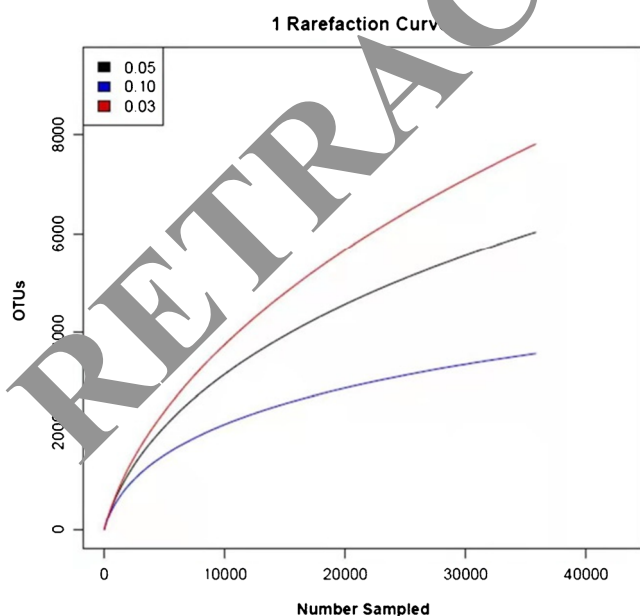


Fig. 10 Dilution curve of sample 1

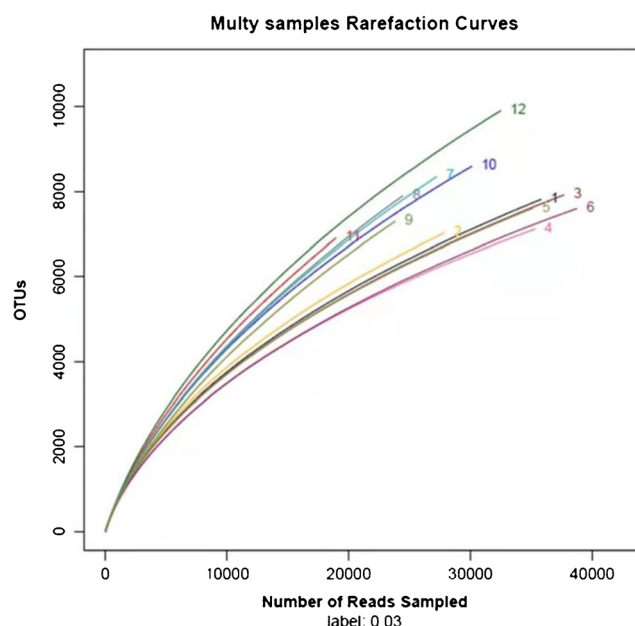


Fig. 11 Dilution curves of all samples

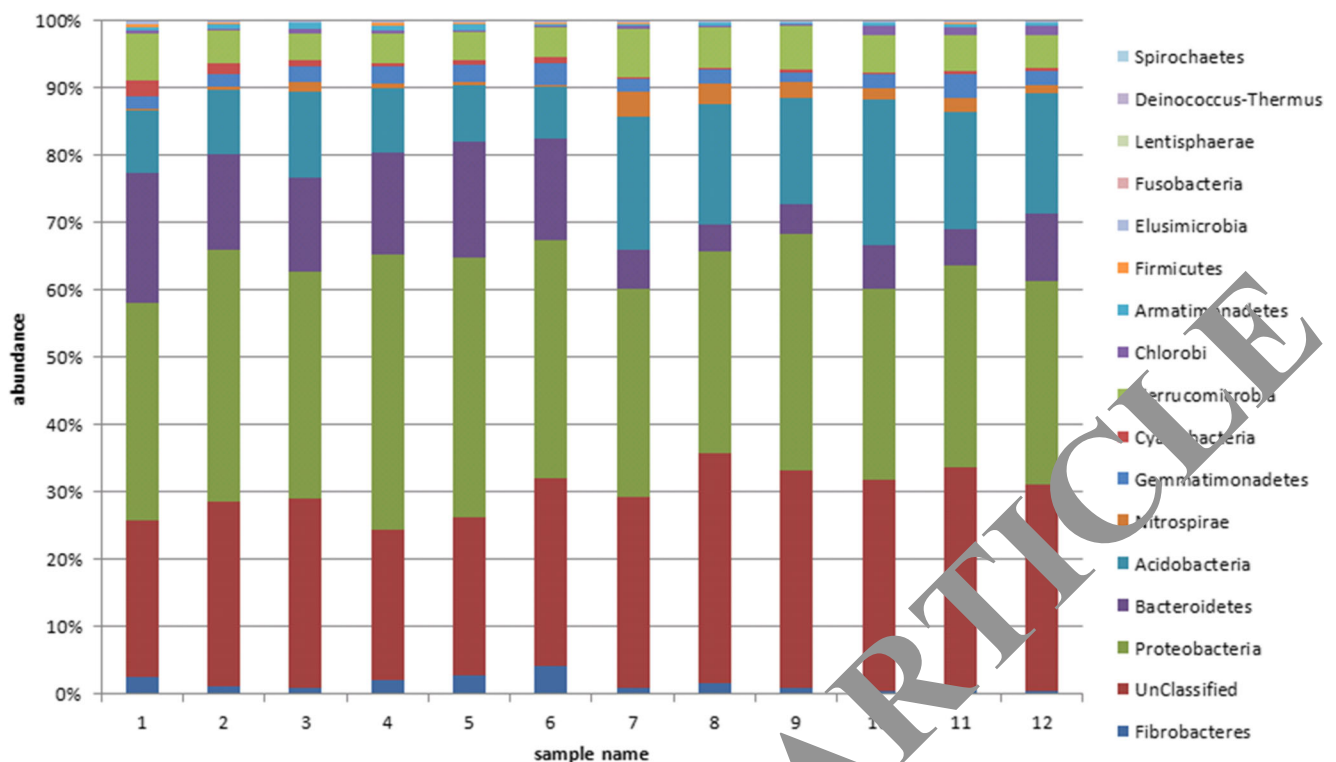


Fig. 12 Differences in community structure composition at each sample gate level

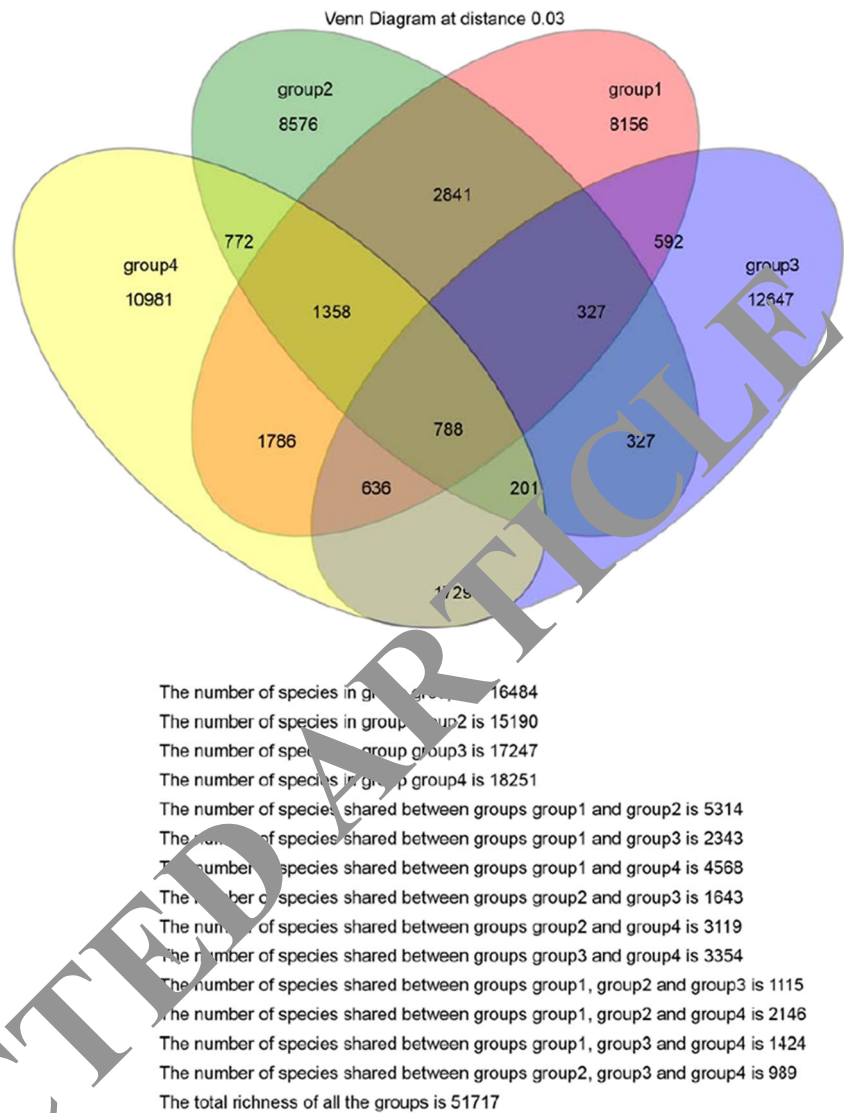
horizontal distribution, the soil enzyme activity is significantly different, and the older the tree, the higher the soil enzyme activity. There is a close relationship between soil enzyme and physical and chemical properties. Huang et al. found that urease and protease play a major role in soil enzymes, and density and total nitrogen content play a major role in physical and chemical properties. Tan Fanglin et al. showed that the white oxidase activity of continuous planting of *Casuarina equisetifolia* in Anzhou, Fujian, decreased with the increase of planting costs, and studied the effect of slash pine on the soil enzyme activity of *Casuarina equisetifolia*. It was found that *Acacia taiwan* may reduce the activity of urease enzyme and increase the activity of the other three enzymes. However, *Slash pine* tends to reduce the activity of soil phosphatase, polyphenol oxidase and peroxidase.

Changes in soil properties in different restoration model of degraded ecosystem

Vegetation is the center of material circulation and energy flow in the ecosystem. Vegetation affects the accumulation and distribution of soil organic matter, and changes the soil microbial clusters and activities through falling matter and roots. Therefore, different land use patterns and vegetation restoration methods have a great impact on soil quality. Wang Gongbo et al. found that the planting patterns of

windbreak forests are different, and the relative soil water content is very different. The planting of multi-layer windbreak forests increases the soil salt concentration and reduces the accumulation of soluble salt in the soil. The soluble salt of the soil in the tree + low planting mode is much lower than that in the low planting mode. The spatial variation of soil nutrients in different windbreak forests is obvious. The multi-level windbreak forest planting model has obvious positive effects on the accumulation of soil nutrients and the improvement of soil fertility, increasing the species in the windbreak forest. The planting mode of mixed forest can increase the biodiversity of windbreak forest and promote the conversion from ineffective nutrients to available nutrients. Su Guangshi analyzed the change characteristics of soil surface microorganisms and enzyme activities under various land use patterns in Chengjiang Karst Watershed, Guangxi Zhuang Autonomous Region. As a result, the total number of soil microorganisms in shrubland and grassland is the largest, followed by natural forest, shrubland, and dry land, and farmland and artificial forestland are the least returned. The activities of tyrosinase, amylase, and protease in natural forests are higher than other land types, and the activities of returned farmland and arid areas are the lowest. Compared with slope farmland, Lu Shiyang et al. studied soil characteristics in 7 common natural land patterns in karst peak clusters. The results show that different models of reducing farmland to forest

Fig. 13 VENN diagram of each sample



and grassland can improve the characteristics of soil microorganisms. Except for the ranging model, all other models can increase the total number of soil microorganisms. The content of soil microbial biomass carbon (MBC) has increased significantly, and the content of soil microbial biomass carbon is the highest on abandoned land. The content of soil microbial biomass nitrogen (MBN) hardly changed, showing a decreasing trend. Various models of farmland returning to forest and grassland have greatly increased the activities of soil urea enzyme and phosphatase. The above research shows that the difference in soil nutrient levels under different vegetation communities will affect the structure and activities of the soil microbial community, and affect the material and energy cycles between plants, microorganisms, and soil. Soil microorganisms carry out the biochemical process of soil nutrient elements. Their cluster structure and activities affect the soil nutrient level and the possibility of nutrient utilization, and secondly affect the overall stability of the ecosystem. It

provides an important foundation for the restoration and reconstruction of fragile ecosystems.

Virtual scene development and optimization technology for VR games

Physical rendering and global illumination technology

In physical rendering, physically based lighting calculations are based on the physical characteristics of light propagation close to the actual lighting conditions, so the physical rendering is particularly true in the performance of natural objects.

Global Illumination (GI for short) is a simulation based on the physical characteristics of light transmission. This is an algorithm used to simulate complex actions, such as simple interaction and bounce, but consumes a lot of memory during the rendering process.

Fig. 14 Thermal map based on the distribution and abundance of each sample

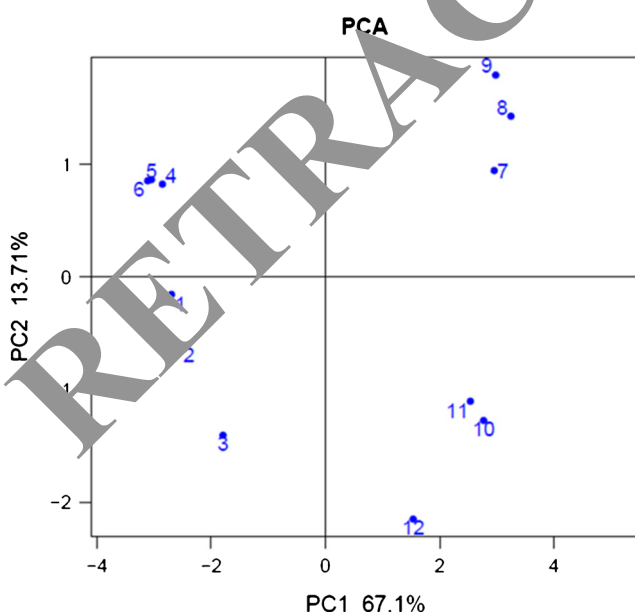
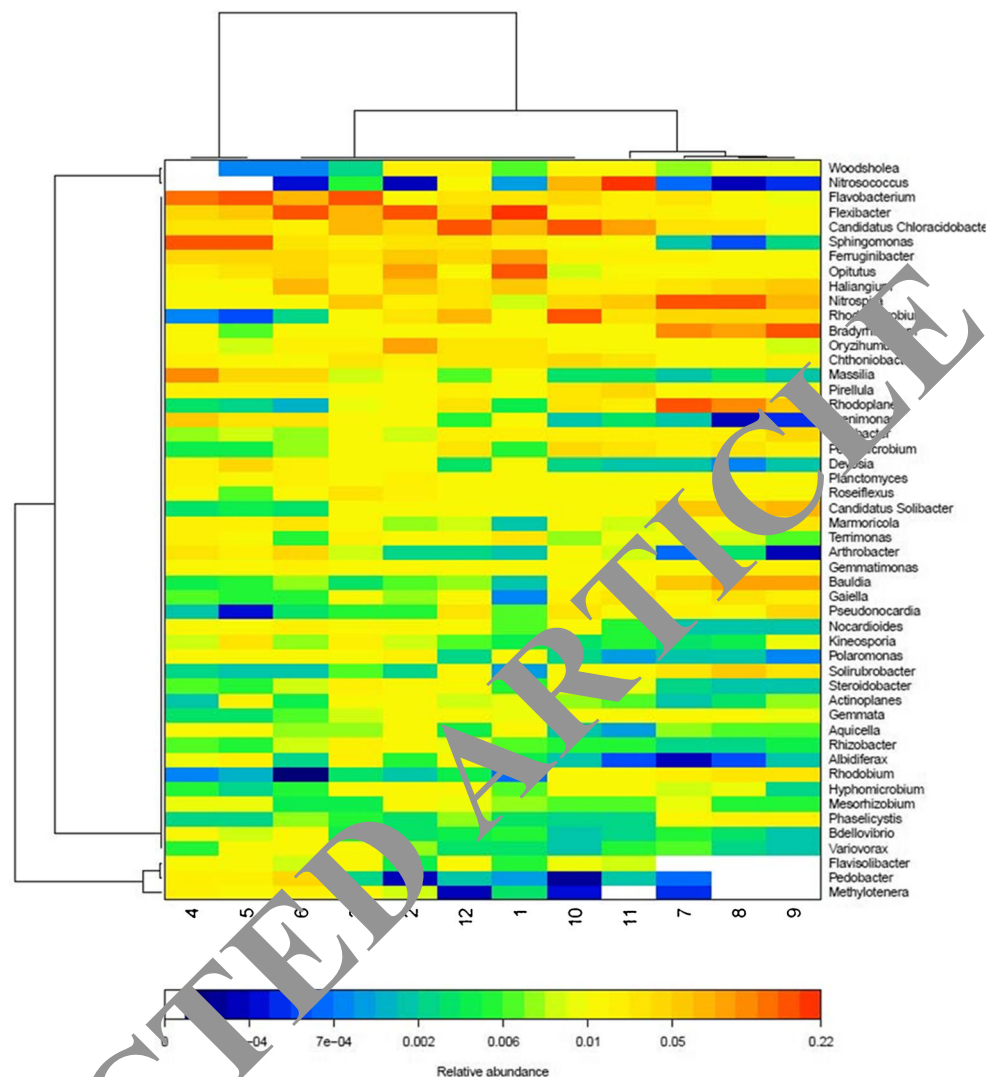


Fig. 15 PCA analysis of the microbial composition between samples

Level of detail technology

Level of detail (LOD) can be divided into technical LOD and masked LOD. The operating principle of the model LOD is to select models with different accuracy according to the distance between the camera and the object. If the LOD of the shader is applied to the whole or sub shaders, and the value of the LOD is less than the set value, the corresponding shader will not work.

Batch processing optimization and GPU Instancing technology

At present, the main technical problems of cross-platform development of games have been solved, and the biggest development difficulty now lies in the insufficient operating capabilities of the CPU and GPU of the mobile device. Batch processing is one of the most common solutions to the

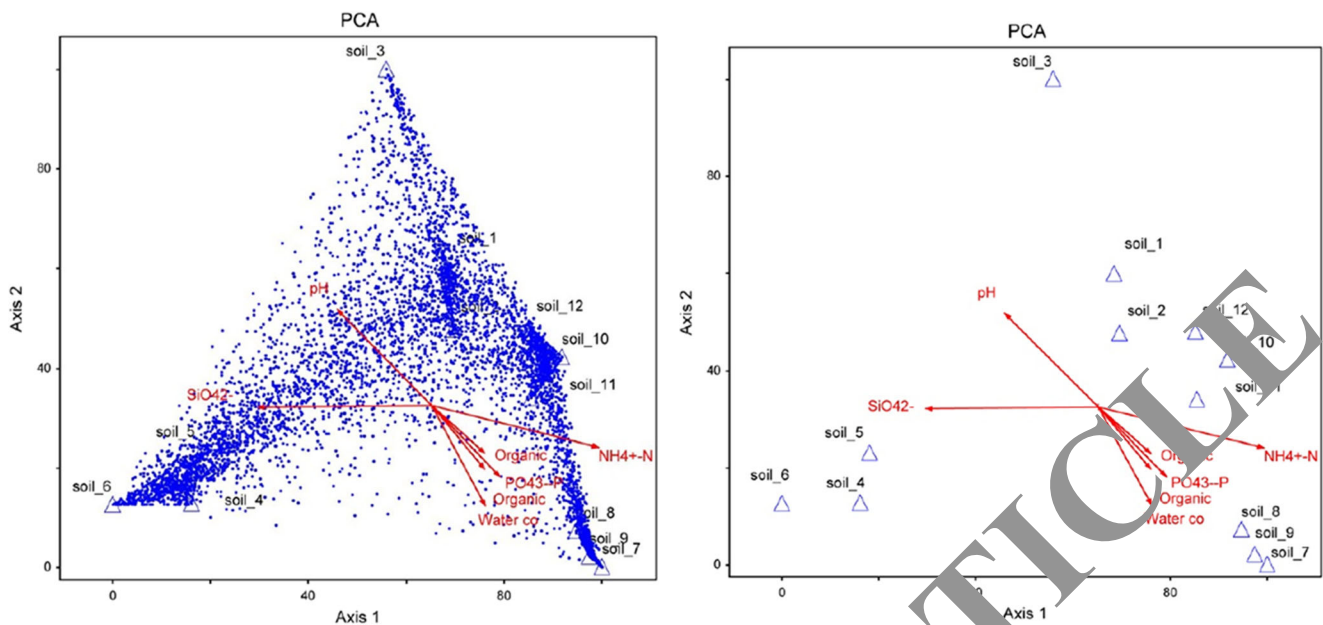


Fig. 16 PCA of each sample and physical and chemical factors

problem of insufficient performance, including StaticBatching and DynamicBatching.

Another good way to improve graphics performance is to use GPUInstancing. The biggest advantage of GPUInstancing is that it can reduce memory usage and CPU overhead. As a result, the efficiency of the model is greatly improved.

Panoramic video technology

As the name suggests, the panoramic image is a 720-degree or 360-degree panoramic image developed and expanded based on 720-degree or 360-degree panoramic technology. The principle of action is to take all-round shots around the center point and connect multiple photos to generate a video with an aspect ratio of 2:1. Finally, through the corresponding player or virtual display device, the audience can observe omnidirectional and three-dimensional without blind spots.

Analysis of influencing factors of VR game experience process

The most commonly used methods to measure flow experience are questionnaire surveys and interviews. A few researchers use physiological indicators to evaluate the flow experience of virtual reality. At present, the flow questionnaire widely recognized at home and abroad is the flow state and flow tendency questionnaire developed by Jackson et al. The dimension of the flow experience questionnaire is mainly developed by Csikszentmihalyi's 9 elements. Research on flow experience needs to confirm the feeling of control, immersion, timely feedback, and fun.

Focusing on VR games includes concentration, presence, and change in time, which means you always focus on games.

The so-called fun refers to the fun that the player has during the VR game, and all the positive emotions that the player can enjoy through the operation.

Analysis of influencing factors of VR game experience results

The flow results of VR games in this study mainly refer to the process of the game and the satisfaction with the results, the expansion of the game experience, etc. It is the result of the player's flow experience during the game. This paper combines the characteristics of VR games and refers to the results of the research so far, using the flow experience result questionnaire survey item. Compared with normal electronic games, VR games have many opportunities to use VR equipment to bring players a realistic game environment and provide players with a rich virtual experience. When the player is immersed in the virtual game environment, people do not notice various emotional experiences including the physical environment, and thus regard the game as a part of reality, and the player experiences a more realistic environment.

Conclusion

The research in this paper shows that the three vegetation restoration models can effectively increase soil nutrients, but the increase in phosphorus is very small (indicating the lack of soil phosphorus), the soil pH value is reduced, and the pure

forest of *Casuarina equisetifolia* significantly reduces the soil (less than 4.00), acidification. After vegetation restoration, the activities of the 5 enzymes behave differently in different sandy bare restoration models. After the bare sandy vegetation was restored, the amount of biomass and soil microorganisms increased significantly. The principal component analysis method was used to comprehensively evaluate the soil quality of various remediation models, including the main factors affecting soil fertility such as soil carbon, nitrogen, and phosphorus.

The motivational factors of VR games have a great influence on the flow experience, the development of skills has a great influence on people, and music has a great influence on the feeling of control. The flow experience of VR games has only a partial impact on the flow results. Control and happiness have a large positive impact on satisfaction, and control and immersion have a positive impact on the expansion of the game experience.

Declarations

Conflict of interest The authors declare that they have no competing interests.

References

- Ali T (2016) Water-hammer control in pressurized-pipe flow using an inline polymeric short-section. *Acta Mech* 227(3):777–791
- Bai Y, Sun L, Wei CH (2020) A coupled gas flow-mechanical damage model and its numerical simulations on high energy gas fracturing. *Geofluid* 2020:1–14. <https://doi.org/10.1155/2020/307037>
- Cuderman JF (1986) Effects of wellbore liquids in propellant-based fracturing. The 27th U.S. Symposium on Rock Mechanics (USRMS), Tuscaloosa, Alabama, June 1986. ARMA 86-0562
- Feng N, Cheng S, Yu H, Shi W, Liu X, Liu Q (2016) The flow and heat transfer characteristics of compressed high-pressure air injection wells. *Arab J Geosci* 11:519. <https://doi.org/10.1007/s12517-018-3878-7>
- Gao H, Wang ZM, Wang J et al (2017) Study on friction coefficient and model of air-propelled control screen. *China Petro Mach* 45(11):12–17
- Han HS (1994) Evaluation on method of calculating compressibility factors of natural gas. *Oil & Gas Stor Trans* 13(1):13–18
- Hossain MM, Rahman MK, Rahman SS (2000) Hydraulic fracture initiation and propagation: roles of wellbore trajectory, perforation and stress regimes. *J Pet Sci Eng* 27:129–149
- Huang BX, Li PF, Ma J, Chen S (2014) Experimental investigation on the basic law of hydraulic fracturing after water pressure control blasting. *Rock Mech Rock Eng* 47(4):1321–1334
- Huang B, Xiong PQ, Sun L (2018) Numerical simulation technology of deflagration fracturing in offshore sandstone reservoir. *China Sci Paper* 13(11):1319–1324
- Jaimes MG, Castillo RD, Mendoza SA (2012) High energy gas fracturing: a technique of hydraulic pre-fracturing to reduce the pressure losses by friction in the near wellbore — a Colombian field application. SPE Latin America and Caribbean Petroleum Engineering Conference, Mexico City, Mexico, April 2012. <https://doi.org/10.2118/152886-MS>
- Khabirov SV (2016) Gas pressure equalization in a porous medium filling a pipe with a closed end under shock loading. *J Appl Mech Tech Phys* 57(6):989–996
- Kutter HK, Fairhurst C (1971) On the fracture process in blasting. *Int J Rock Mech Min Sci Geomech Abstr* 8(3):181–202
- Lanari M, Fakhimi A (2015) Numerical study of contributions of shock wave and gas penetration toward induced rock damage during blasting. *Compu Part Mech* 2(2):197–208
- Li D, Zhao G, Yang WY (1995) High energy gas fracturing (HEGF): mechanism and practice. International Meeting on Petroleum Engineering, Beijing, China, November 1995. <https://doi.org/10.2118/29992-MS>
- Liu GH, Zhao X (2018) Fracture mechanism experiment of top control fluid in compound perforation. *J China Univ Petro* 32(6):88–91
- Liu GZ, Yang Z, Zhang S et al (2018) Analysis on the application of HEGF in oilfield. *Petro Chem Equip* 21(7):33–38
- Ma GW, An XM (2008) Numerical simulation of blasting-induced rock fractures. *Int J Rock Mech Min Sci* 45(6):966–975
- Majidi G, Soheil M, Ahmad J (2015) Numerical analysis of rock fracturing by gas pressure using the extended finite element method. *Pet Sci* 12(2):304–315
- Pasamaneck RS (1993) Controlled propellant fracturing for vertical wells. SPE-26271-MS
- Rao KVN, Brinsden MS, Gilliat J, et al (2014) World's first deepwater propellant perforation for a depleted carbonate subsea gas well in Malampaya, 14–16 October 14. APOG Adelaide, Australia. SPE-171527-MS
- Rossmann HP, Knasmüllner RE, Daehnke A et al (1996) Wave propagation, damage evolution, and dynamic fracture extension. Part II. *Blasting Mater Sci* 32(4):403–410
- Sahil M, Peggy R, Alicia S (2017) Experimental investigation of propellant fracturing in a large sandstone block. Offshore Technology Conference, Houston, Texas, USA, May 2017. <https://doi.org/10.4043/27563-MS>
- Sun ZY, Liu CY, Su JZ et al (2010) Dynamic analysis of the fracture propagation by multi-pulse HEGF. *J Southwest Petrol Univ (Sci Technol Ed)* 32(6):121–124
- Sun L, Yang WY, Yi F et al (2017) Feasibility research on liner-completion deflagration fracturing technique. *Spec Oil Gas Reser* 24(4):161–165